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Jacob D Krol

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EDUCATION

- Ph.D. Computer Science** | University of Colorado Boulder | Boulder, CO Aug 2024 — present
- Advisor: Ryan Layer
 - GPA: 4.00/4.00
- Graduate, Non-degree** | University of Colorado Anschutz Medical Campus | Aurora, CO Sep 2023 — Dec 2023
- GPA: 4.00/4.00
 - BIOS 7747: Machine Learning for Biomedical Applications, graduate course offered by the Colorado School of Public Health
- Bachelor of Science, Neuroscience** | Michigan State University | East Lansing, MI Sep 2020 — Aug 2022
- GPA: 3.89/4.00
 - Graduation awards: *Cum Laude*
 - Semester awards: Dean's List
- Math and Science Transfer Program** | Washtenaw Community College | Ann Arbor, MI Sep 2017 — Apr 2020
- GPA: 3.52/4.00
 - Semester awards: Honor Roll

RESEARCH INTERESTS

Computational biology, Statistics, Population genomics, Algorithms, Natural language processing, Graph theory, History, Evolution

RESEARCH EXPERIENCE

- Graduate Research Assistant** Aug 2024 — Present
- Layer lab | Department of Computer Science | PI: Ryan Layer
University of Colorado Boulder | Boulder, CO
- Designing an algorithm to find gene fusion driver mutations in human cancer from population scale DNA and RNA sequencing datasets.
- NIH/NLM Data Science and Informatics Scholars Summer Intern** Jun 2024 — Aug 2024
- Protein and Genome Evolution Research Group | PI: Aravind Iyer
National Institute of Health (NIH/NLM/NCBI) | North Bethesda, MD
- Developed methods for discovering genomic conflict systems through natural language processing (NLP) and network analysis.
 - Trained Word2Vec model to learn context-aware gene embeddings and extracted genomic conflict systems from cosine similarity networks of the gene embedding space.
 - Identified co-localized genes/proteins N-grams ($N \geq 2$) associated with biological conflict systems across the evolutionary tree using entropy and frequency analysis.
 - Presented findings at two poster events on NIH campus.
 - Member of Transformers for AI journal club.
- Information Sciences Professional** Nov 2022 — May 2024
- JRavi Lab | Department of Biomedical Informatics | PI: Janani Ravi
University of Colorado School Anschutz School of Medicine | Aurora, CO
- Developed an internal R package for machine learning analysis of antimicrobial resistant (AMR) bacterial pathogens.
 - Full stack web development for *MolEvolvR: a web app for characterizing proteins using molecular evolution and phylogeny.*, Krol, et al., 2023; bioRxiv, DOI: doi.org/10.1101/2022.02.18.461833 (jrvilab.org/molevolvr).
 - Submitted graduate-level fellowships to national agencies: 1) NSF GRFP and 2) DOE CSGF.
 - Presented research talks and software development workshops at international conferences: 1) R Bioconductor 2023 and 2) Great Lakes Bioinformatics 2023.
 - Led workshops on shell programming, Git version control, remote linux computing, and Docker.
 - Managed the lab's GitHub organization: code review plus managing issues and pull requests.
 - Peer-mentored Ph.D. (3) and undergraduate students (3) on R/Python/Shell programming, version control, data wrangling, developing ML models, hypothesis testing, web development, and presenting scientific research.
 - Assisted in public deployment of R Shiny dashboard for the publication *The Phage-shock-protein (PSP) Envelope Stress Response: Discovery of Novel Partners and Evolutionary History*. DOI: doi.org/10.1101/2020.09.24.301986
 - Performed system administrator tasks (e.g., server onboarding, dependency/user/data/resource management) for lab's server.

Student Research Assistant

May 2022 — Nov 2022

Krishnan Lab & Malmstrom Lab, Department of Computational Mathematics Science and Engineering (CMSE)
Michigan State University, East Lansing, MI

- Developed machine learning classifiers trained on viral omics data to predict plant virus host and virus taxonomy.
- Analyzed the association of protein domains with viral host phenotype through Fisher's Exact test hypothesis testing
- Extracted latent viral genomic features with principal component analysis (PCA).
- Presented the approach and results in project meetings.

PUBLICATIONS

1. **Krol, J. D. et al.** MolEvolvR: A web-app for characterizing proteins using molecular evolution and phylogeny. *bioRxiv*. DOI: doi.org/10.1101/2022.02.18.461833 (2023).
2. Ghosh, A., Brenner, E. P., Vang, C. K., Wolfe, E. P., Burke, J. T., Lesiyon, R. L., Manpearl, K. R., Sridhar, V., **Krol, Jacob D, et al.** From sequence to signature: Uncovering multiscale AMR features across bacterial pathogens with supervised machine learning. *bioRxiv*. <https://www.biorxiv.org/content/early/2025/07/07/2025.07.03.663053> (2025).

AWARDS AND FUNDING

- **Awarded** University of Colorado Graduate School Student Travel Grant 2025
- **Awarded** National Institute of Health (NIH) Intramural Research Training Award (IRTA) Summer Student Traineeship. 2024
- **Awarded** NSF ACCESS explore prSoject allocation for high performance computing: "Processing sequences into feature datasets for microbial genotype to phenotype machine learning." 2024
- Submitted Winter 2023 Department of Energy Computational Science Graduate Fellowship (DOE CSGF): 2023
A multi-modal deep learning technique predict antibiotic resistance via computational chemistry and bacterial genomics. (Not awarded).
- Submitted Fall 2023 NSF Graduate Research Fellowship Program (GRFP): 2023
"Microbial phenotype prediction with graph machine learning methods". (Not awarded).

PRESENTATIONS

Research and technical talks

- R/Bioconductor: Cancer and Evolution track. *MolevolvR a web-app for protein characterization.* Jul 2023
Boston University, Boston, MA.
- Great Lakes Bioinformatics Conference. *MolEvolvR a web-app for protein characterization.* May 2023
McGill University, Montreal, CA.
- Great Lakes Bioinformatics Conference. *How and when to build a web-app or R 2023 package?* May 2023
McGill University, Montreal, CA.

Posters

- National Institute of Health Summer Poster Day *The language of genomic conflict systems* Aug 2024
National Institute of Health, Bethesda, MD.
- American Society for Microbiology Rocky Mountain Branch *Classifying antimicrobial resistance in high-impact pathogens* Apr 2024
University of Colorado Boulder, Boulder, CO.
- CU Department of Biomedical Informatics Annual retreat. *Classifying antimicrobial resistance in high-impact pathogens* Sep 2023
University of Colorado Anschutz, Aurora, CO.

PEER MENTEES

PhD Students

- Charmie Vang, Biomedical Sciences program, CU Anschutz 2023
- Keenan Manpearl, Computational Bioscience program, CU Anschutz 2023
- Jill Bilodeaux, Microbiology program, CU Anschutz 2023

Undergraduates

- Skylar Stefanowicz, B.S. Biology, Metropolitan State University of Denver 2024
- Ethan Wolfe, B.S. Biochemistry & Molecular Biology with CMSE and additional minors, Michigan State University 2022 — 2024

REFERENCES

- **Ryan Layer, Ph.D.**; ryan.layer@colorado.edu
— Assistant professor, University of Colorado Boulder, Department of Computer Science
- **Aravind Iyer, Ph.D.**; aravind@ncbi.nlm.nih.gov
— Senior investigator of Protein and Genome Evolution Research Group, NIH/NLM/NCBI